Global Epidemiological Parameters Database

Survey Debrief

World Health Organization | HUB
Pandemic and Epidemic Intelligence

Aim:

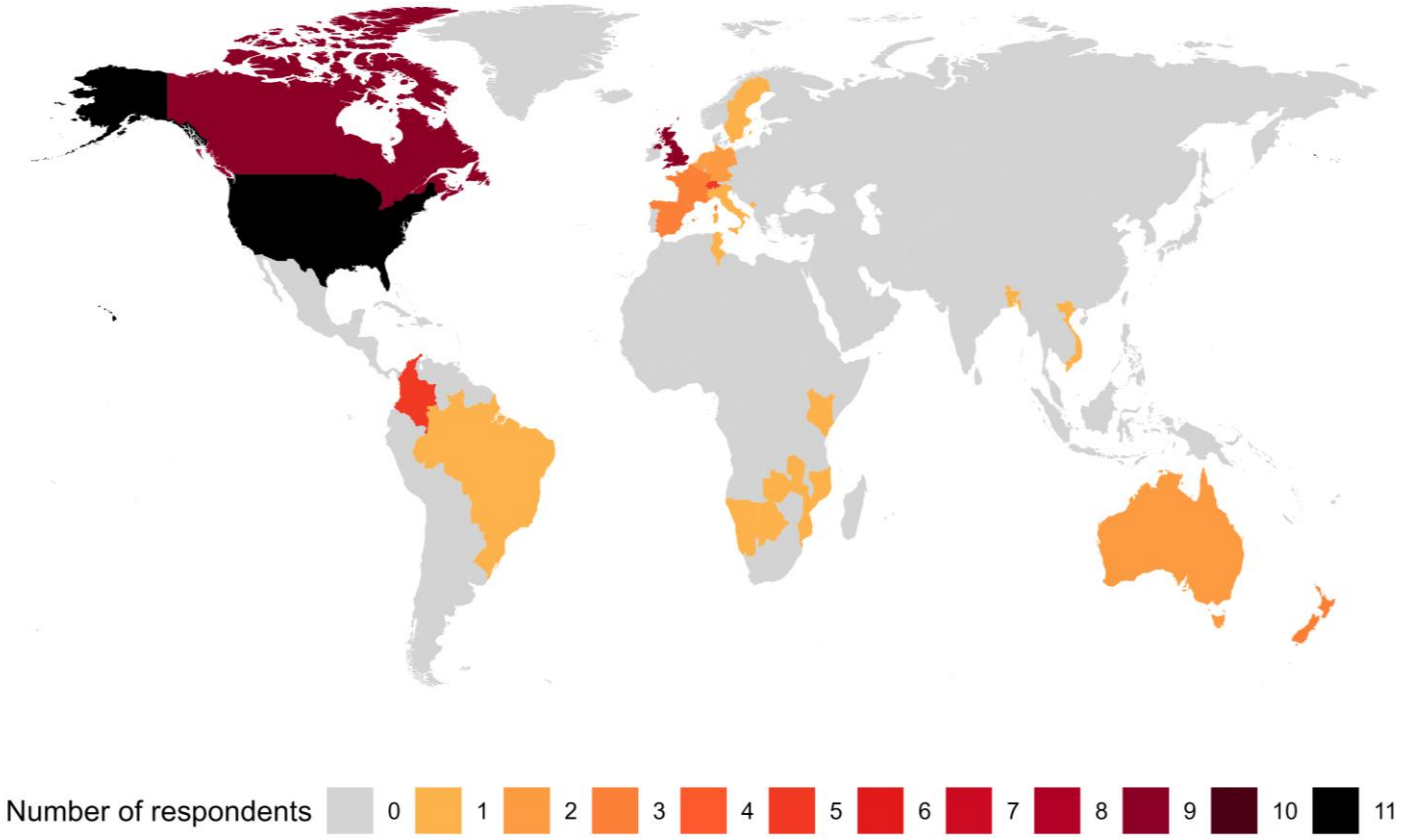
Consolidate database structure in accordance with community perspectives and needs

Structure:

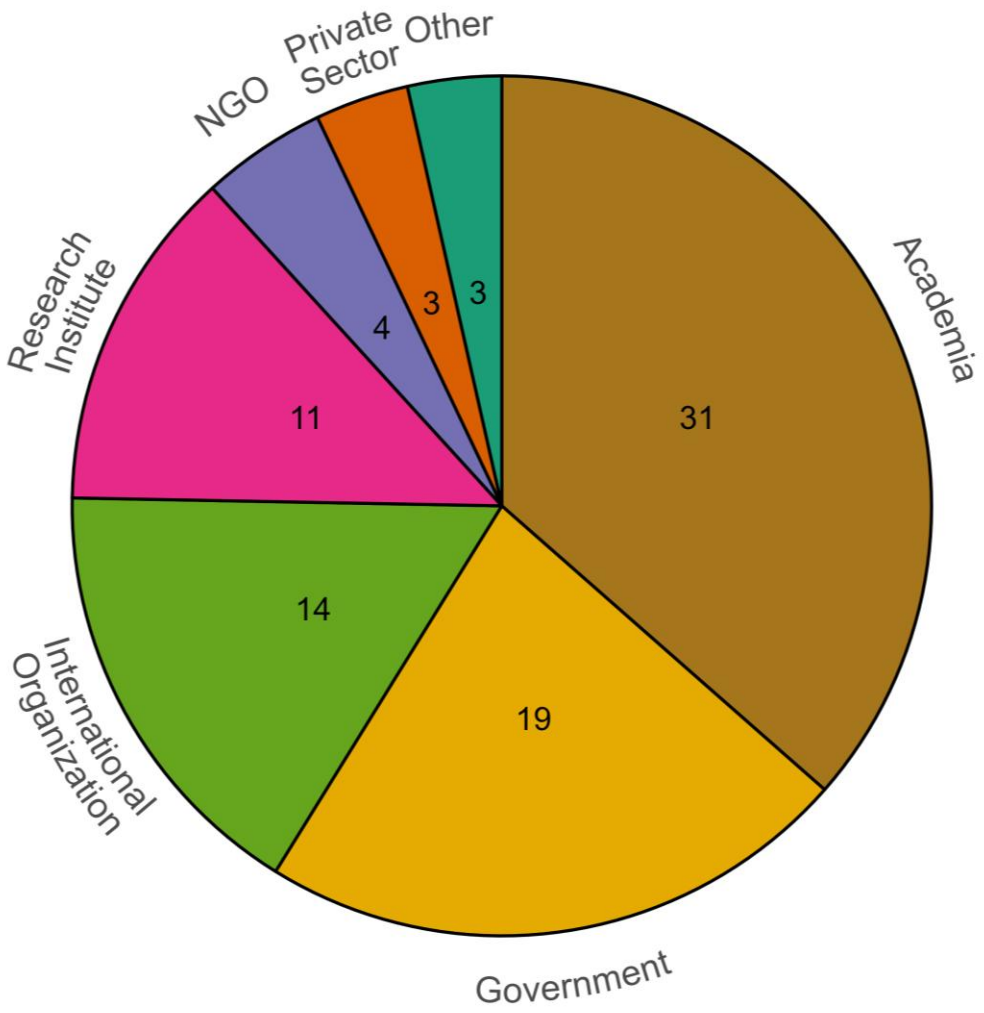
1. Overview of survey participants - who is the “community”?
2. Survey results, section by section
   a) Summary of results and feedback
   b) Brief discussion of key questions and issues
3. Concluding discussion

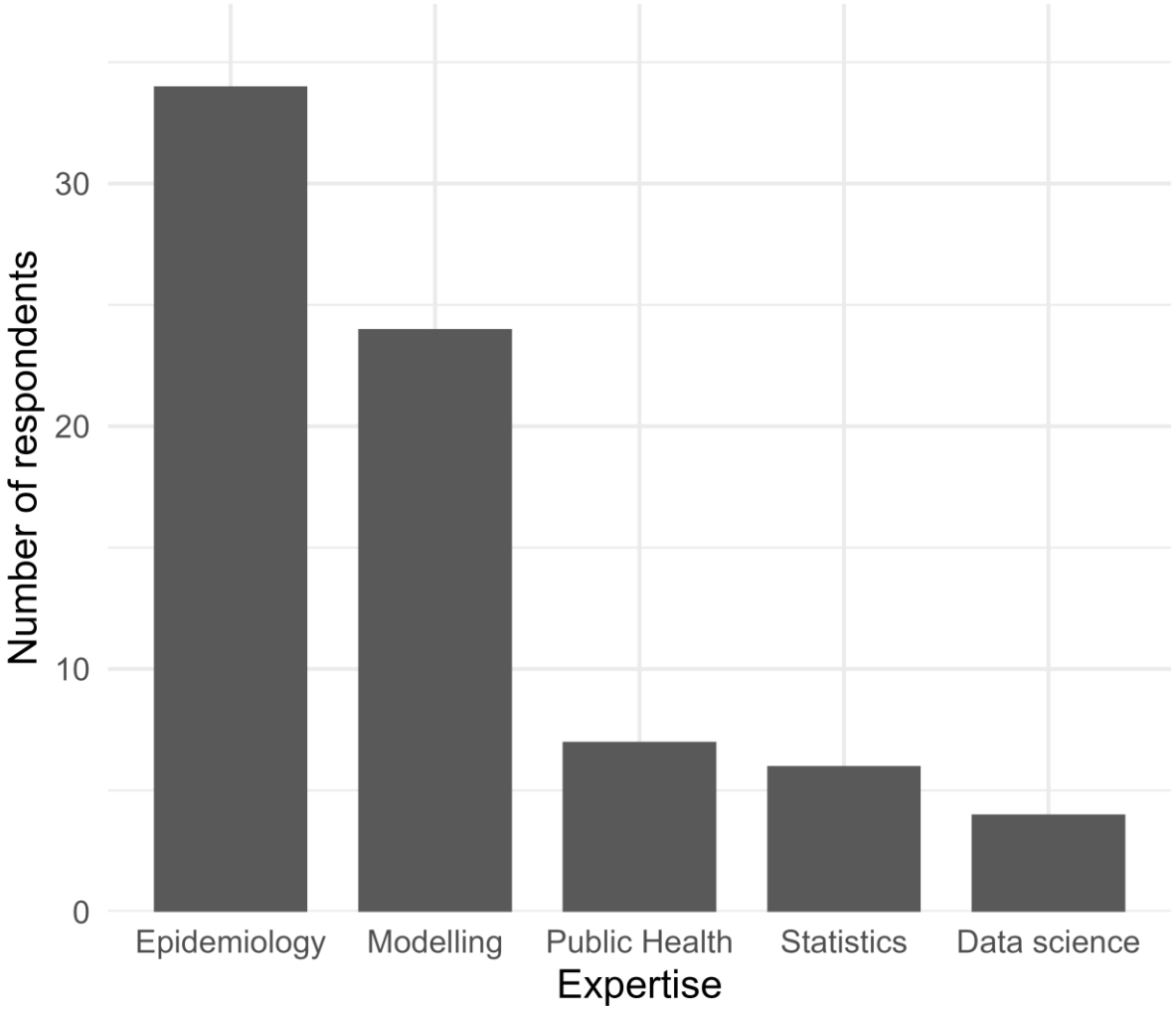
Section 1: Survey Participants

86 respondents from 25 countries and 53 Institutions



A variety of sectors and expertise





Section 2: Survey Results

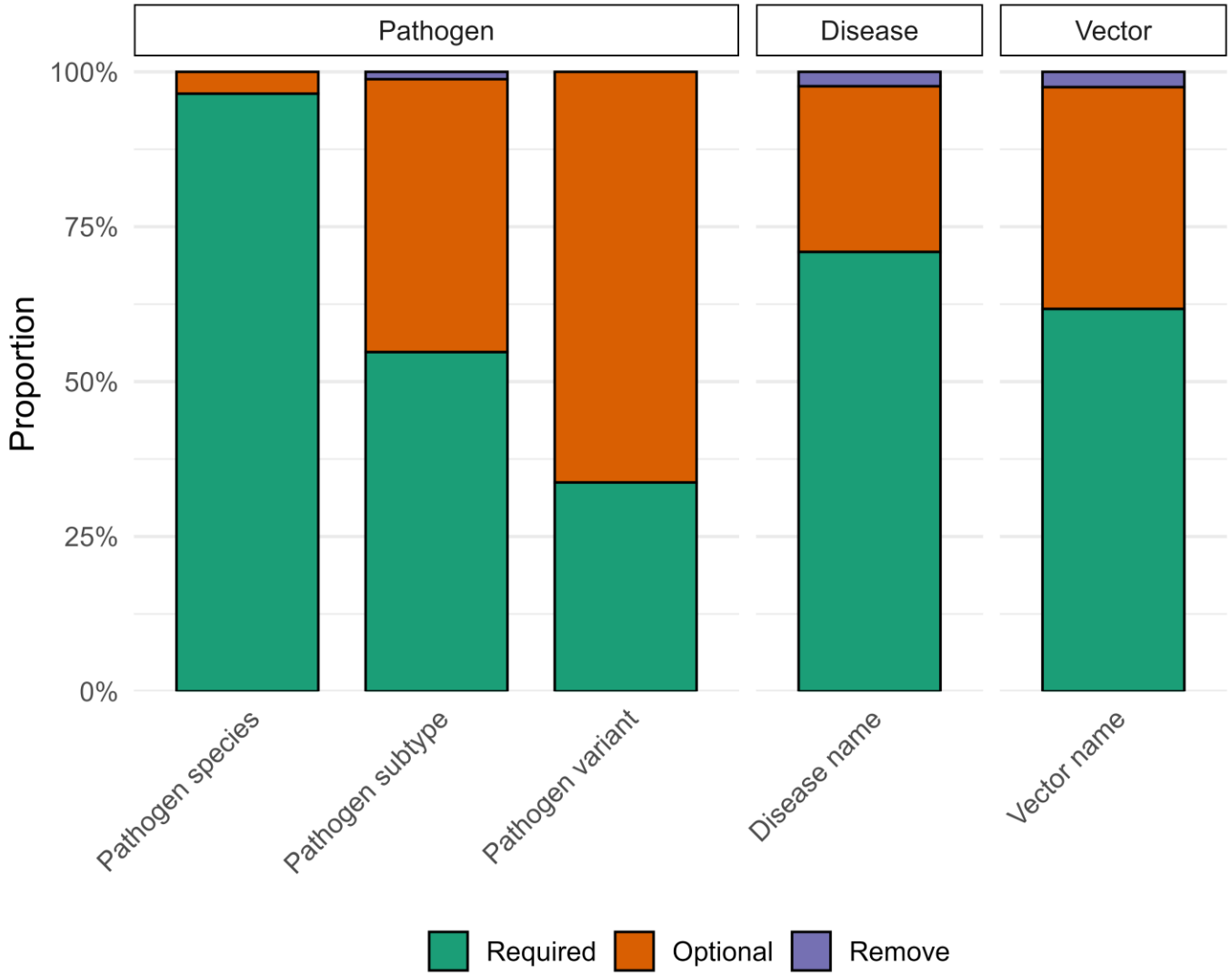
Pathogen, disease and vector designations

Results & Feedback

- Importance of subtypes, variants depends on pathogen (e.g. influenza vs measles)
- Higher-level classifications like pathogen family could be useful (e.g. flavivirus, alphavirus)
- Automated field-completion if pathogen species is provided?

Discussion

- Should pathogen species be selected from a list or free-text?



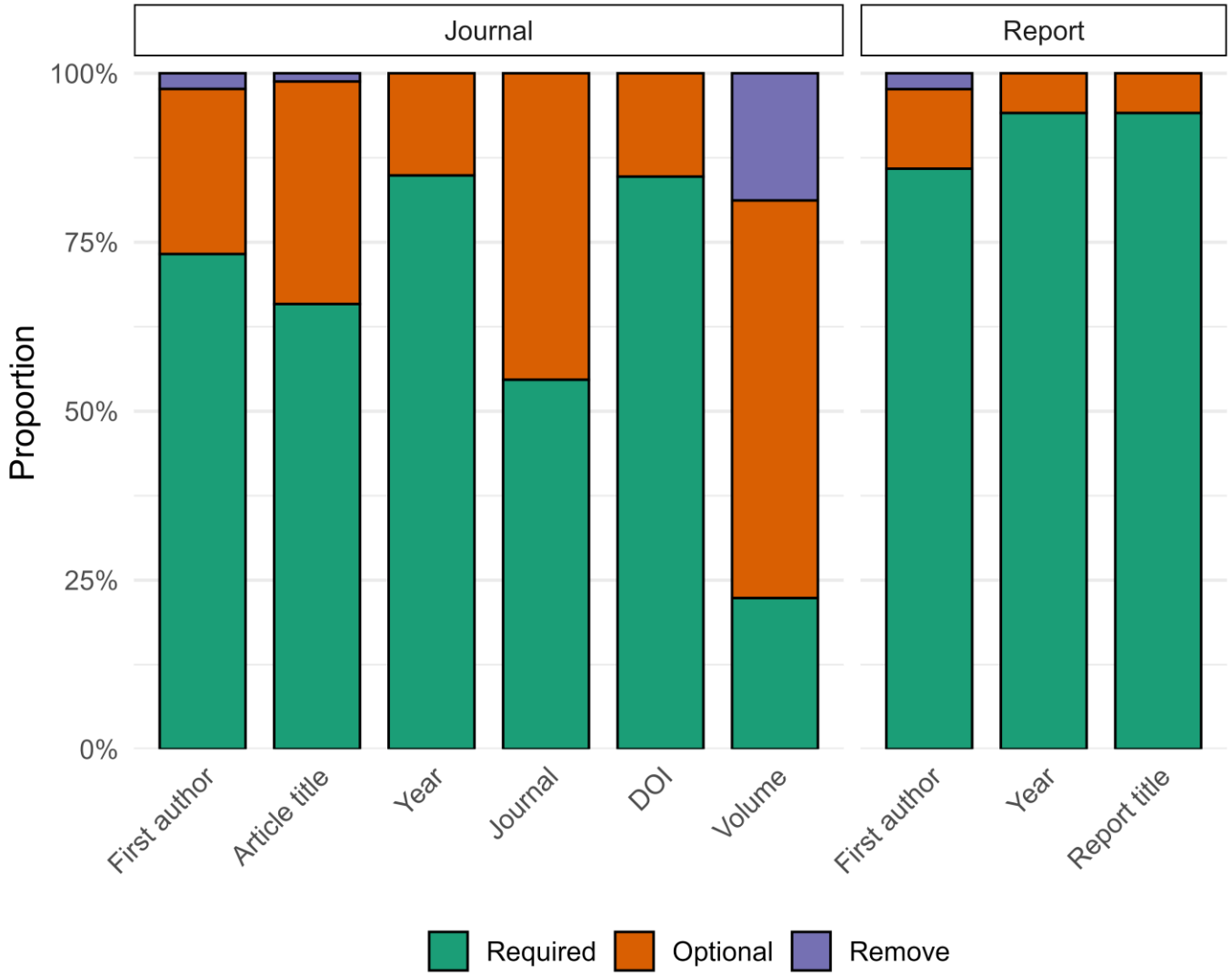
Source

Results & Feedback

- Unique identifier (UID) needed: DOI for article and URL for unpublished report
- Automatically generate citations from UID?

Discussion

- Should everything besides UID be optional?



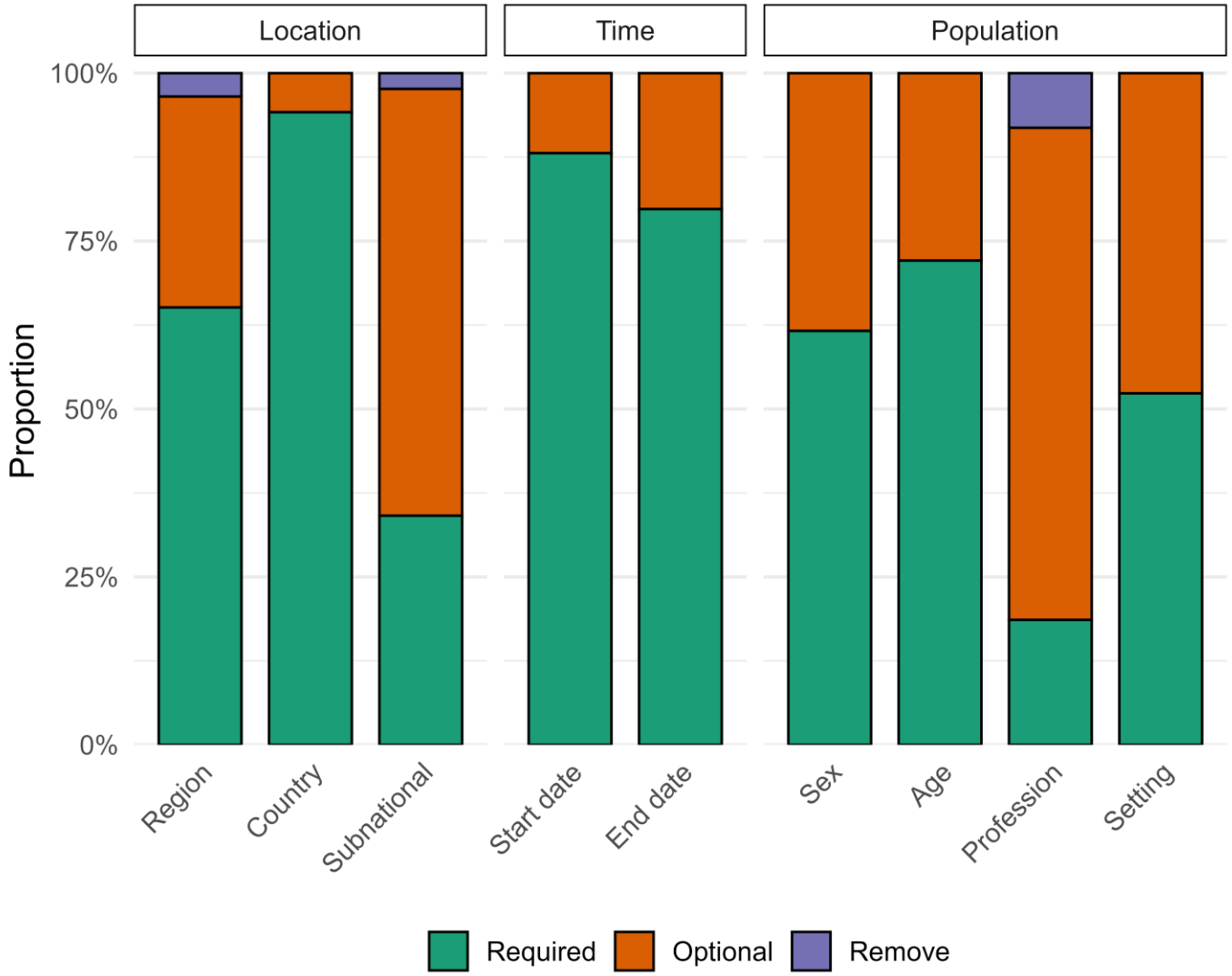
Context

Results & Feedback

- Basic location and time indicators considered necessary by most respondents
- Context is important for some parameter types (e.g. transmission rates highly context specific)
- Population data needed for disaggregated parameters
- Large number of optional population covariates that **vary between parameters and studies** and cannot be specified clearly ahead of time

Discussion

- What does the start date refer to?
- Should population be included? As a free-text field?
- Should “population” be used for characterising representativeness of the study population or for disaggregating parameter estimates?



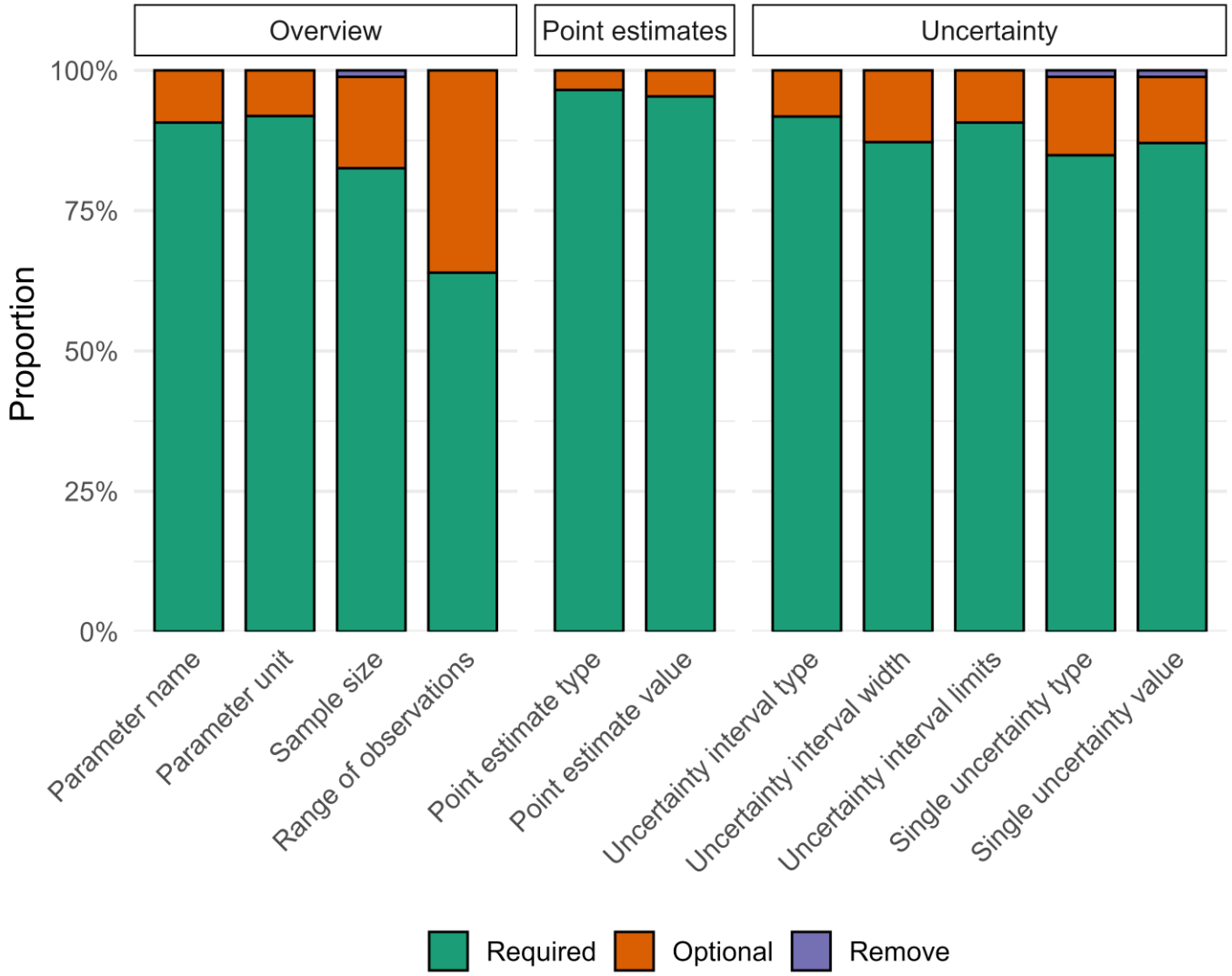
Parameter estimates

Results & Feedback

- Clear parameter definitions needed
- Uncertainty listed as a “required” field by >90% of participants

Discussion

- Should a measure of uncertainty be necessary for submission?
- Can/should parameter names be selected from pre-specified list?



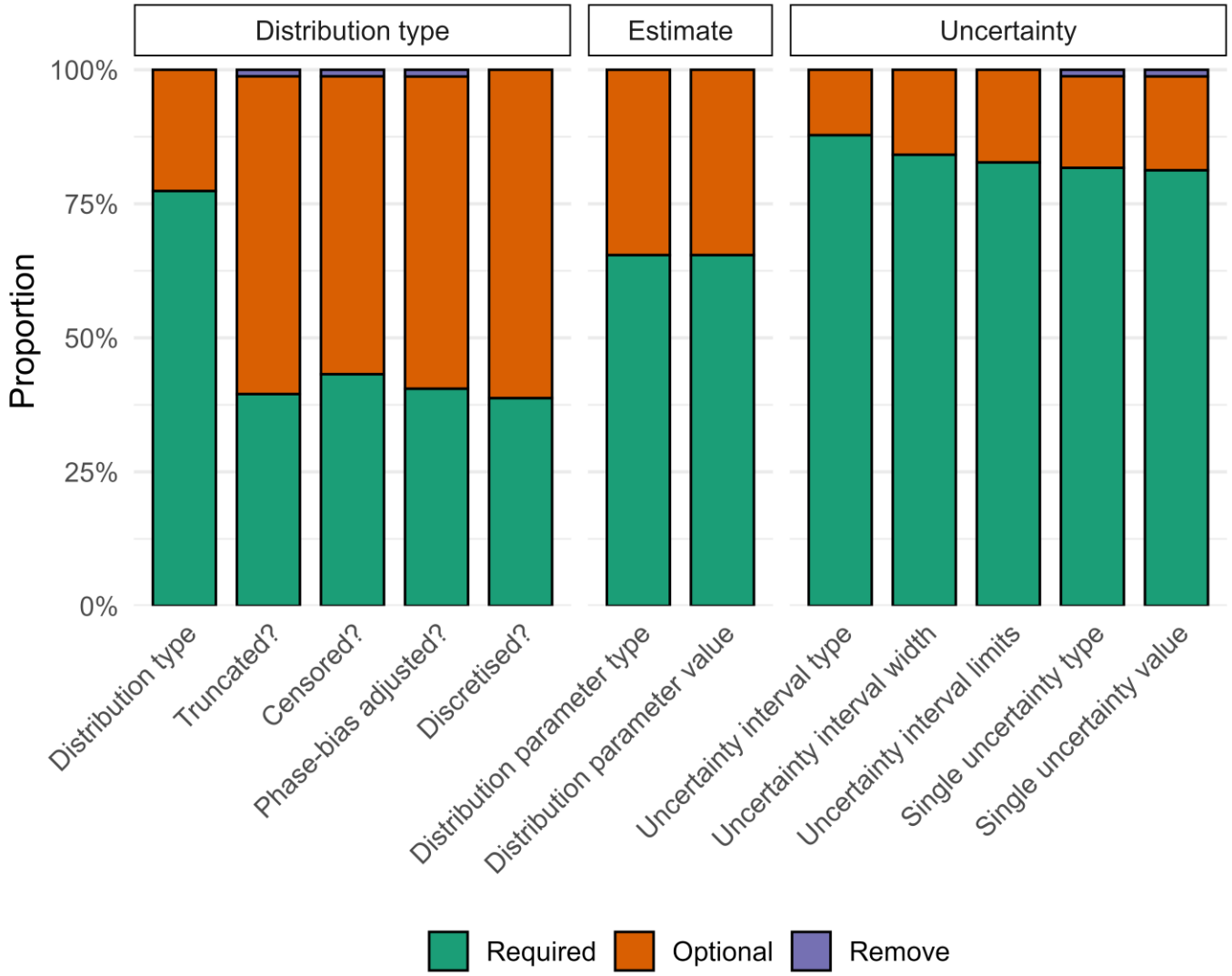
Distribution estimates

Results & Feedback

- At a minimum: **central estimate, variance and distribution type**. Parameters can be calculated from these if necessary.
- How to deal with non-parametric distributions? Provide the empirical distribution and calculate summary statistics?
- Parameterizations of distributions differ a lot. Include a reference list of distributions and parameterizations?

Discussion

- Are central estimate, variance and distribution type a valid minimum submission?
- Is uncertainty around the distribution parameters necessary for submission?
- Can empirical distributions be uploaded to the database or only summary statistics?



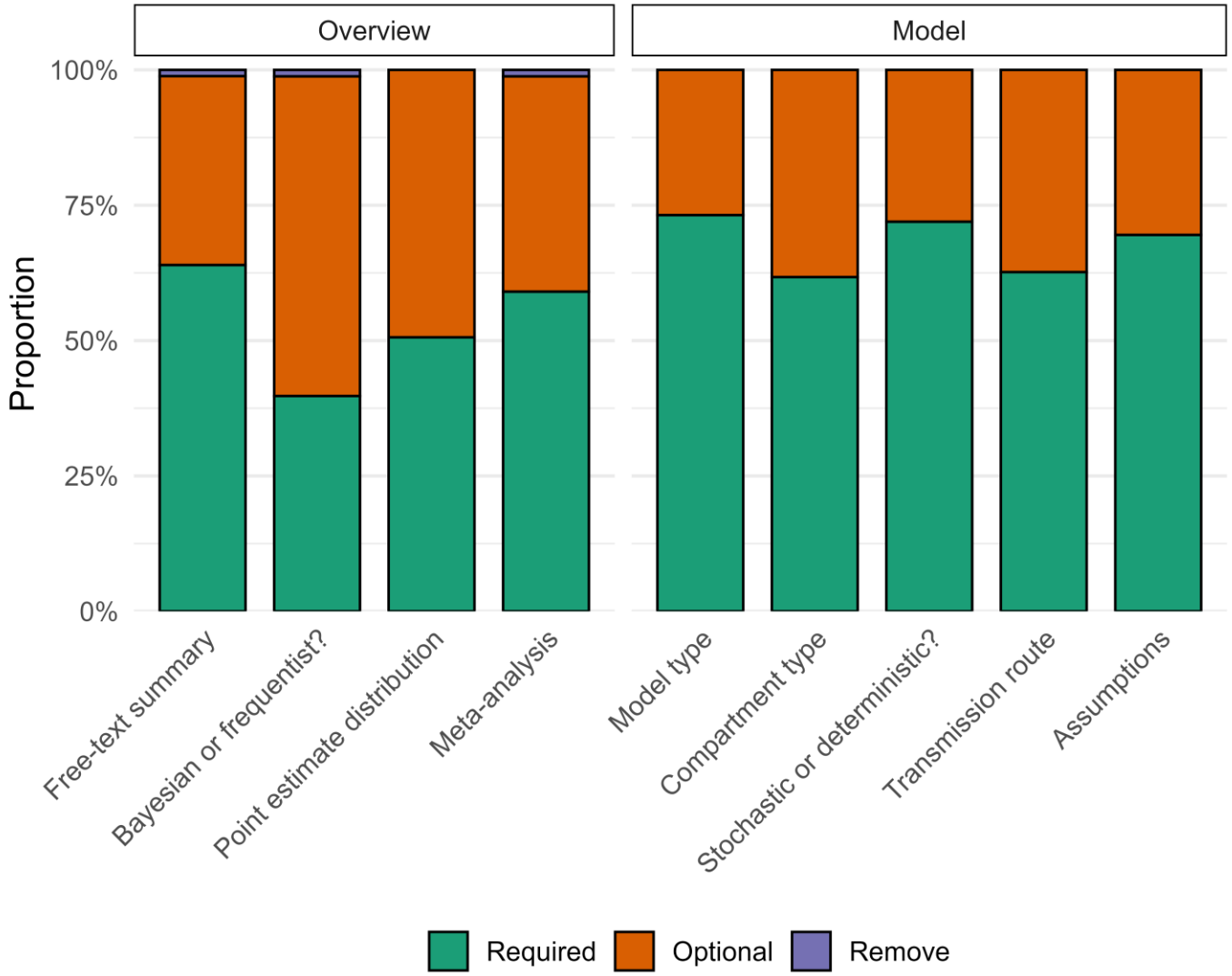
Methods

Results & Feedback

- Specify study type (e.g. descriptive, observational study)
- "Models have unique features that cannot be specified according to predefined categories: simple categorization may be misleading."
- "The ultimate reference must be the original paper. Unclear how useful it would be to collect these details in free-text fields, unless you can decide upon a short list of valid values."
- "Meta-analysis requires a lot more information to be extracted. I am not sure meta-analyses have a place in this database."

Discussion

- Methods: free-text field, free-text with sections, drop-down list or reference to original paper?
- Should meta-analyses be included in the database?



Section 3: Concluding discussion

Concluding discussion

Results & Feedback

- “Not everything needs to be in the database”
- “Database structure should drive best practice, it’s better to have a small database of high-quality information than a large database that is flooded with submissions that are hard or risky to interpret”
- “What about sensitive data? We should get thinking about people's locations, privacy and equity.”

Discussion

- Emphasis on structured fields, free text or minimal submissions with links to source?
- Emphasis on quality or quantity?